```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import pearsonr

# Data
data = {
    'Age': [12, 15, 16, 20, 21, 24, 25, 30],
    'Costume Type': ['Spooky', 'Funny', 'Spooky', 'Spooky', 'Spooky', 'Funny', 'Spooky', 'Fu
}
df = pd.DataFrame(data)

# Convert costume type to binary (Spooky=0, Funny=1)
df['Costume Code'] = df['Costume Type'].map({'Spooky': 0, 'Funny': 1})

# Observed correlation
observed_corr, observed_p = pearsonr(df['Age'], df['Costume Code'])

# Permutation test
np.random.seed(42)
n_permutations = 10000
permuted_corrs = []

for _ in range(n_permutations):
    shuffled_codes = np.random.permutation(df['Costume Code'])
    perm_corr = df['Age'].corr(pd.Series(shuffled_codes))
    permuted_corrs.append(perm_corr)

# Calculate p-value (one-tailed, positive correlation)
p_value = np.mean(np.array(permuted_corrs) >= observed_corr)

fig, ax2 = plt.subplots(figsize=(15, 6))

# Plot 1: Permutation test results
ax2.hist(permuted_corrs, bins=50, alpha=0.7, color='gray',
         density=True, edgecolor='black', linewidth=0.5)
ax2.axvline(observed_corr, color='red', linestyle='--', linewidth=2,
           label=f'Observed Correlation = {observed_corr:.3f}')
ax2.set_xlabel('Correlation Coefficient')
ax2.set_ylabel('Density')
ax2.set_title(f'Permutation Test Results\np-value = {p_value:.4f}')
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ax2.legend()
ax2.grid(alpha=0.3)

plt.tight_layout()
plt.show()

# Additional visualization: Age distribution by costume type
plt.figure(figsize=(10, 6))

# Box plot
plt.subplot(1, 2, 1)
box_data = [df[df['Costume Type'] == 'Spooky']['Age'],
           df[df['Costume Type'] == 'Funny']['Age']]
plt.boxplot(box_data, tick_labels=['Spooky', 'Funny'])
plt.ylabel('Age')
plt.title('Age Distribution by Costume Type')
plt.grid(alpha=0.3)

# Print results
print("DATA SUMMARY:")
print(df)
print(f"\nSPOOKY costumes (n={sum(df['Costume Code'] == 0)}):")
print(f"  Mean age: {df[df['Costume Code'] == 0]['Age'].mean():.1f}")
print(f"  Ages: {list(df[df['Costume Code'] == 0]['Age'])}")
print(f"\nFUNNY costumes (n={sum(df['Costume Code'] == 1)}):")
print(f"  Mean age: {df[df['Costume Code'] == 1]['Age'].mean():.1f}")
print(f"  Ages: {list(df[df['Costume Code'] == 1]['Age'])}")
print(f"\nSTATISTICAL RESULTS:")
print(f"Observed Correlation: {observed_corr:.4f}")
print(f"P-value (one-tailed): {p_value:.4f}")
print(f"Interpretation: {'SUPPORTS' if p_value < 0.05 else 'DOES NOT SUPPORT'} Mr. Draculous
```
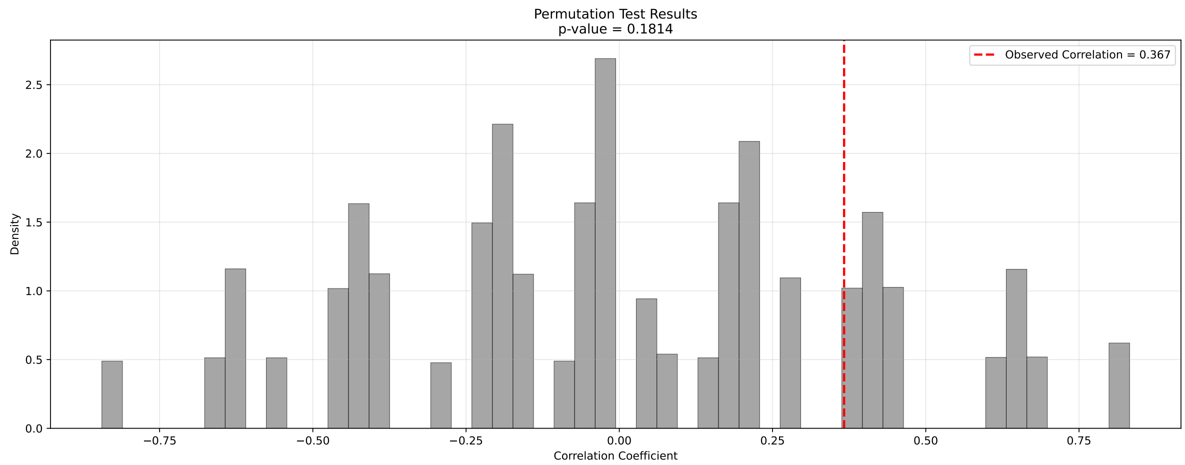
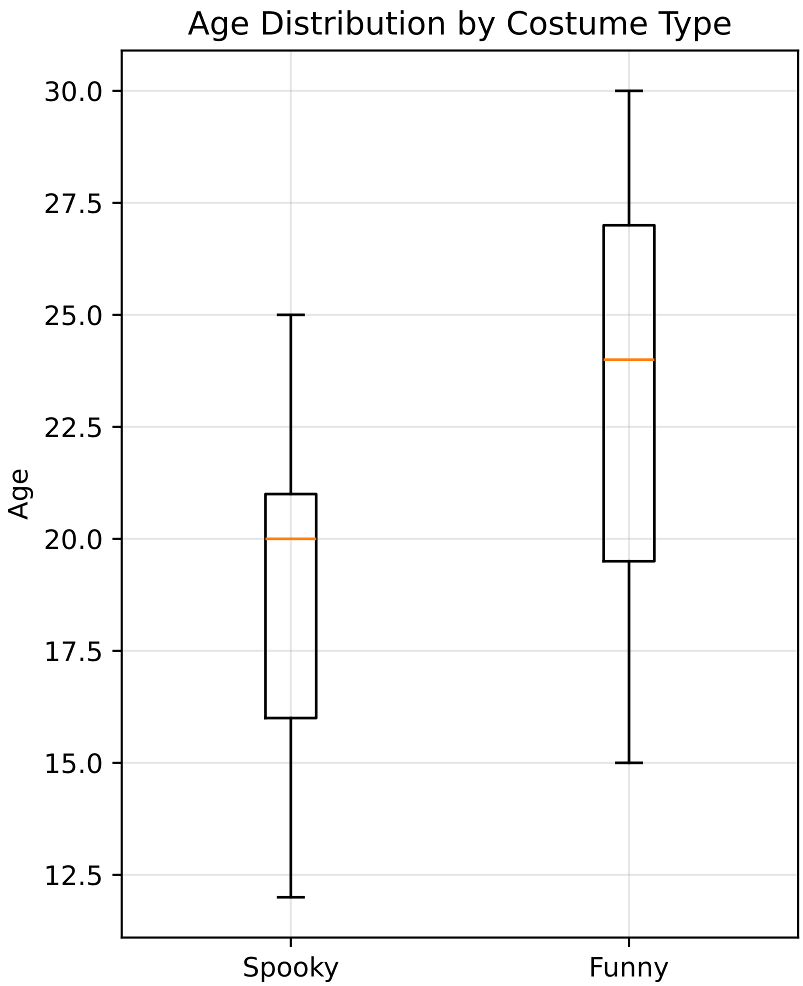




```
DATA SUMMARY:
   Age Costume Type  Costume Code
0   12       Spooky             0
1   15        Funny             1
2   16       Spooky             0
3   20       Spooky             0
4   21       Spooky             0
5   24        Funny             1
6   25       Spooky             0
7   30        Funny             1

SPOOKY costumes (n=5):
  Mean age: 18.8
  Ages: [12, 16, 20, 21, 25]

FUNNY costumes (n=3):
  Mean age: 23.0
  Ages: [15, 24, 30]

STATISTICAL RESULTS:
Observed Correlation: 0.3668
P-value (one-tailed): 0.1814
Interpretation: DOES NOT SUPPORT Mr. Draculous' hypothesis
```


Age Distribution by Costume Type
30.0
27.5
25.0
22.5
20.0
17.5
15.0
12.5
Age
Spooky
Funny